

Register Number: \_\_\_\_\_

**Amrita Vishwa Vidyapeetham**  
**Amrita School of Computing, Bangalore**  
**Department of Computer Science and Engineering**  
**Periodical Test 3, December 2022**  
**19CSE353 Mining of Massive Datasets**

Class: VII CSE A and B

Duration: 2 hrs

Answer all the questions

Date: 05.12.2022

Max. Marks: 50

| CO   | Course Outcome                                                                                                                                                                                    |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CO01 | Understand the importance of data to business decisions, strategy and behavior. Predictive analytics, data mining and machine learning as tools give new methods for analyzing massive data sets. |
| CO02 | Expose the students to Big data systems like Hadoop, Spark and Hive.                                                                                                                              |
| CO03 | Explore means to deal with huge document databases and infinite streams of data to mining large social networks and web graphs. Also, learn Algorithms suitable for large scale mining.           |
| CO04 | Use of analytic tools, case-studies to provide first hand insight into how big-data problems and their solutions are commercially derived.                                                        |
| CO05 | Design Large Scale Machine Learning algorithms with practical hands-on experience for analyzing very large amounts of data.                                                                       |

**Answer all questions**

- 1.A A business wishes to segment its customers into a small number of groups so that it can effectively target marketing efforts at different customer types. Discuss the process and the order of the steps. How is Customer segmentation done in Datamining? [2 + 2] [BTL 4] [CO3]
- 1.B Is Web a directed graph or undirected graph? Discuss the two challenges of web search with examples. [2 + 4] [BTL 2] [CO3]
- 2.A Are machine learning models statistical models? Justify with an example. Why is Statistics important to Machine Learning, discuss with an example? [3 + 3] [BTL 3] [CO2]
- 2.B When should log transformation be used? Explain it's significance in mining massive datasets. [2 + 2] [BTL 2] [CO2]
- 3.A Is Association Rule Mining a type of supervised learning or unsupervised learning technique? Justify. Delineate the Metrics used to measure the associations between data items using necessary equations. [2 + 2] [BTL 2] [CO3]
- 3.B When we search on the internet, we want to see the most relevant pages. Discuss the relevant algorithm used to determine the pages which are more authoritative on the internet based on their popularity to ensure users see pages that are most likely to be of use to them. Discuss the challenges of web search. [4 + 2] [BTL 3] [CO 2]

4.A What is MapReduce? Illustrate a simple example of the working of MapReduce.

[2 + 2]

[BTL 1]  
[CO1]

4.B Consider a Transaction database 'D' below considering the minimum support count 2 and minimum confidence 50%.

[3 + 3]

[BTL 3]  
[CO2]

| TID | ITEMS |
|-----|-------|
| 1   | 2,3   |
| 2   | 1,3,5 |
| 3   | 1,2,4 |
| 4   | 2,3   |

Database D

1. Find all frequent itemsets, and
2. Generate strong association rules from the frequent itemsets using Aprori Algorithm.

5.A What are Data Streams? Discuss the problems associated with Data Streams.

[2 + 6]

[BTL 1]  
[CO1]

5.B How do you mine social network news feeds?

[2]

[BTL 2]  
[CO3]

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Course Outcome / Bloom's Taxonomy Level (BTL) Mark Distribution Table

| CO   | Marks | BTL   | Marks |
|------|-------|-------|-------|
| CO01 | 12    | BTL 1 | 12    |
| CO02 | 22    | BTL 2 | 16    |
| CO03 | 16    | BTL 3 | 18    |
| CO04 | 0     | BTL 4 | 04    |
| CO05 | 0     | BTL 5 | 0     |